

Software Requirement Specifications

<MediPredict AI>



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Meeting Details

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Summary

MediPredict AI will be a web-based machine learning system developed to predict the risk of three major diseases; **Heart Disease, Diabetes, and Kidney Disease**. The system will aim to support early detection by analyzing basic patient health data and providing intelligent risk assessments.

The user will enter medical attributes such as blood pressure, cholesterol, glucose level, creatinine, age, and other relevant health indicators through a single dynamic input form. Based on this input, the system will process the data using three separate **Random Forest classifiers**, each of which will be trained specifically for one disease.

Instead of providing a simple binary output (disease present or not), the system will generate a risk probability score (0–100%) for each disease. These probabilities will be categorized into **these risk levels**:

- 0–10% → Low Risk (No immediate concern)
- 11–20% → Medium Risk (Monitor condition)
- 21–30% → Slightly High Risk (Early warning stage)
- 41–60% → High Risk (Medical consultation recommended)
- 61–80% → Very High Risk (Immediate action required)
- 81–100% → Critical Risk (Urgent medical attention)

This approach will provide a more realistic and informative medical prediction.

All three models will run simultaneously on the same patient data, similar to how doctors evaluate multiple possible conditions from a single set of symptoms. This will make the system a **multi-disease prediction framework** rather than a single-disease classifier.

The frontend of the system will be developed using **Streamlit**, which will provide an interactive and user-friendly interface. The backend will be implemented using **Python**, along with machine learning libraries such as Scikit-learn, Pandas, NumPy, and Joblib.

Patient records will be stored in a **MySQL database**, including input features, predicted risks, disease type, and prediction date. This will enable history tracking and future analysis.

The system will be developed using tools such as **Visual Studio Code, Jupyter Notebook, and GitHub** for coding, experimentation, and version control.

Overall, MediPredict AI will be designed as an intelligent healthcare assistant that will help in early disease risk detection by integrating multiple machine learning models into a single, efficient screening system.

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1. Introduction

MediPredict AI is a web-based machine learning system designed to predict the risk of Heart Disease, Diabetes, and Kidney Disease. It uses a single patient input form to collect health data such as blood pressure, glucose, cholesterol, and creatinine levels. The system processes this data using three separate Random Forest models to generate risk predictions for each disease. Results are shown as Low, Medium, or High risk based on probability scores. The main goal is to support early disease detection and improve health awareness.

1.1. Purpose

The purpose of *MediPredict AI* is to develop an intelligent web-based system that can predict the risk levels of **Heart Disease, Diabetes, and Kidney Disease** using machine learning. The main objective is to support early disease detection by analyzing patient health data and providing a risk-based output (Low, Medium, High) instead of a simple yes/no prediction. This helps users and healthcare providers understand potential health risks at an early stage.

1.2. Scope

The system allows users to enter medical information through a single dynamic form and receive predictions for three diseases simultaneously. The project includes:

- Data input form for patient health parameters
- Machine learning-based prediction models
- Risk classification system
- Patient record storage and history tracking

Not included in this system:

- Real-time hospital integration
- Doctor prescription system
- Advanced medical diagnosis or treatment recommendations
- Emergency response or alert services

1.3 Product Perspective

MediPredict AI is designed as a standalone web-based healthcare prediction system. It acts as a supportive tool that assists in early screening of diseases. The system takes patient input data and processes it through trained machine learning models. The output is displayed on a dashboard, making it easy for users to understand their health risk levels. It can later be integrated into hospital systems for broader use.

1.4. User Characteristics

The system is designed for the following users:

- **Patients / General Users:** Enter their health data and view risk predictions
- **Administrators:** Manage system data, monitor records, and view analytics
- **Healthcare Professionals (optional users):** Use predictions as supporting information for early diagnosis
- **Data Viewers:** Access patient history and reports

Users do not need technical knowledge; the interface is simple and user-friendly.

1.5. Similar apps and systems / Literature Review

Several existing systems focus on single-disease prediction such as:

- Heart disease prediction systems using ML classifiers
- Diabetes prediction models based on glucose and BMI
- Kidney disease detection tools using medical datasets

*However, most of these systems are limited to **only one disease at a time**. Some advanced healthcare platforms provide multiple disease screening but are often complex or not user-friendly. Existing solutions lack a unified interface that predicts multiple diseases from a single input form.*

However, most of these systems are limited to **only one disease at a time**. Some advanced healthcare platforms provide multiple disease screening but are often complex or not user-friendly. Existing solutions lack a unified interface that predicts multiple diseases from a single input form.

MediPredict AI improves upon these systems by:

- Combining three disease predictions in one platform
- Using a single input form for all models
- Providing risk-based classification instead of binary output
- Offering a simple and interactive dashboard

1.6 Proposed Technologies

The system will be developed using the following technologies:

- **Frontend:** Streamlit (interactive web interface)
- **Backend:** Python, Flask / Streamlit logic
- **Machine Learning:** Scikit-learn, NumPy, Pandas, Joblib
- **Database:** MySQL / SQLite
- **Development Tools:** Visual Studio Code, Jupyter Notebook
- **Version Control:** GitHub

2. Requirements

The MediPredict AI system provides core functionality for early disease risk prediction using machine learning. The system allows users to enter their basic medical information through a single form and receive instant predictions for multiple diseases including Heart Disease, Diabetes, and Kidney Disease. The application processes the input data using trained machine learning models and generates risk levels in the form of Low, Medium, and High categories. Additionally, the system stores patient history in a database so that previous predictions can be viewed and analyzed over time. The overall goal of the system is to provide an easy-to-use healthcare screening tool that supports early detection and health awareness.

2.1 Functional Requirements

2.1.1. Patient Health Data Input

- **Name:** FR001
- **Purpose:** To collect patient health information for disease prediction.
- **User(s):** Patient / End User
- **Input:**
 1. Age
 2. Gender
 3. Blood Pressure
 4. BMI
 5. Glucose Level
 6. Insulin Level
 7. Chest Pain (Yes/No)
 8. Cholesterol Level
 9. Creatinine Level
 10. Hemoglobin
 11. Smoking Status (Yes/No)
 12. Family History of Disease

- **Output:** Structured data sent to the prediction models
- **Description:** The system provides a single dynamic form where users enter all required health details. This data is used as input for all three disease prediction models.

2.1.2 Disease Risk Prediction

- **Name:** FR002
- **Purpose:** To predict risk levels for multiple diseases using machine learning models.
- **User(s):** Patient / System
- **Input:** Patient health data from FR001
- **Output:** Risk percentage and category (Low, Medium, High) for each disease
- **Description:** The system uses three separate Random Forest models to analyze input data and generate risk predictions for Heart Disease, Diabetes, and Kidney Disease simultaneously.

2.1.3 Risk Classification

- **Name:** FR003
- **Purpose:** To convert model probability into risk categories.
- **User(s):** System
- **Input:** Probability score (0–100%) from ML models
- **Output:**
 - ✧ 0–10% → Low Risk (No immediate concern)
 - ✧ 11–20% → Medium Risk (Monitor condition)
 - ✧ 21–30% → Slightly High Risk (Early warning stage)
 - ✧ 41–60% → High Risk (Medical consultation recommended)
 - ✧ 61–80% → Very High Risk (Immediate action required)
 - ✧ 81–100% → Critical Risk (Urgent medical attention)

Description: The system categorizes predicted probabilities into meaningful risk levels for easy understanding by users.

2.1.4 Patient Record Storage

- **Name:** FR004
- **Purpose:** To store patient data and prediction history.
- **User(s):** System Admin / Backend
- **Input:** Patient details and prediction results
- **Output:** Stored record in database
- **Description:** All patient inputs and prediction results are saved in a database (MySQL) with timestamps for future reference and analysis.

2.1.5 View Prediction History

- **Name:** FR005
- **Purpose:** To allow users/admin to view past predictions.
- **User(s):** Admin / Authorized User
- **Input:** Patient ID or filter (date/disease type)
- **Output:** List of previous prediction records
- **Description:** The system provides a history view where users can track previous health predictions and analyze changes over time.

2.2 Non-Functional Requirements

- Fast response time for predictions
- Simple and user-friendly interface
- Secure handling of patient data
- Reliable and accurate results
- High system availability
- Compatible with web browsers and devices
- Scalable for future expansion
- Clear and understandable output display
- Handles invalid input safely
- Efficient database storage and retrieval

3. Use Cases and Flow of Processes

Use cases represent the interaction between the user (actor) and the system based on the functional requirements. In the MediPredict AI system, the main actor is the **Patient/User**, who provides health data and receives disease risk predictions. The system processes the input using machine learning models and returns results in the form of risk levels for multiple diseases. The overall system focuses on a single workflow where user input is analyzed and results are displayed instantly.

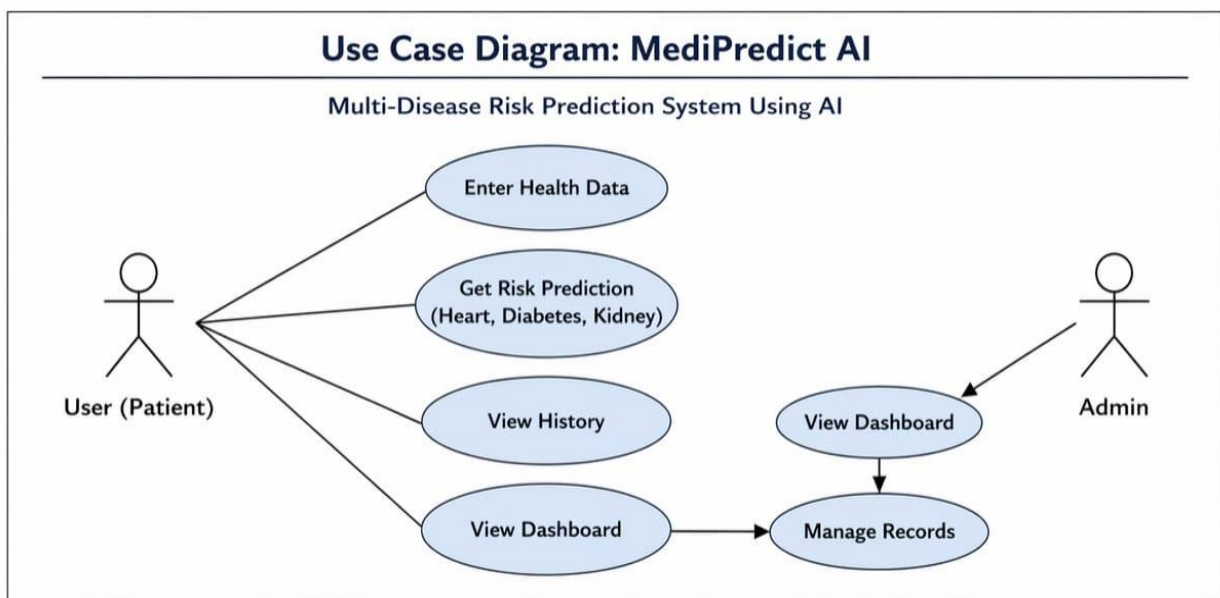
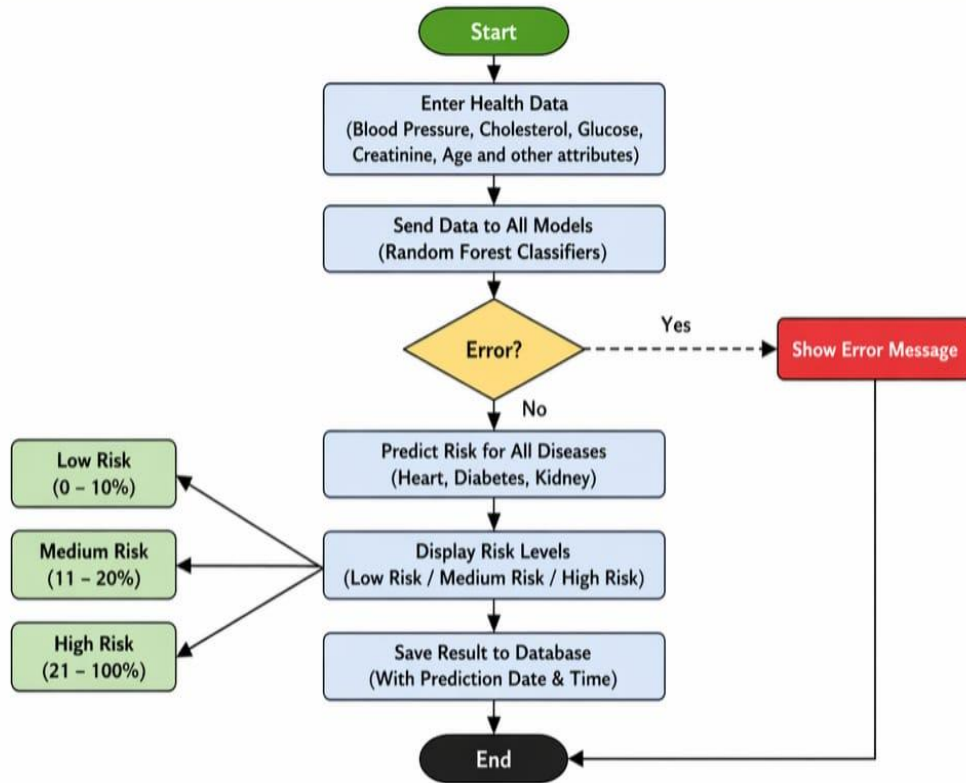


Figure 1: System Level Use Case Diagram

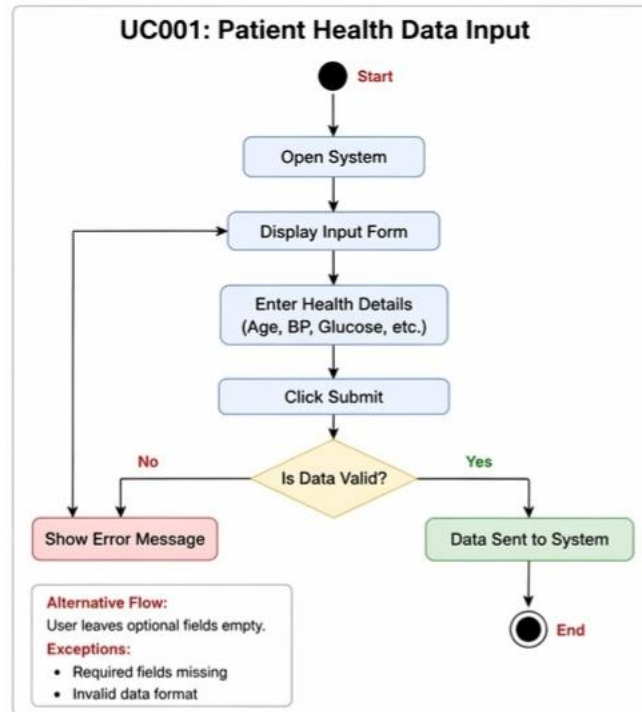
Activity Diagram: Multi-Disease Risk Prediction



3.1.1 Use Case 1

ID	UC001
Name	Patient Health Data Input
Description	This use case describes how the user enters health data into the system.
Requirement(s)	FR001
Actor(s)	Patient / User
Precondition	System must be running and accessible
Postcondition	Patient data is submitted to the system
Basic Flow	<i>Basic Flow</i> 1. User opens the system 2. System displays input form 3. User enters health details (age, BP, glucose, etc.) 4. User clicks submit 5. Data is sent to the system Alternative Flow ● User leaves optional fields empty Exceptions ● Required fields missing ● Invalid data format

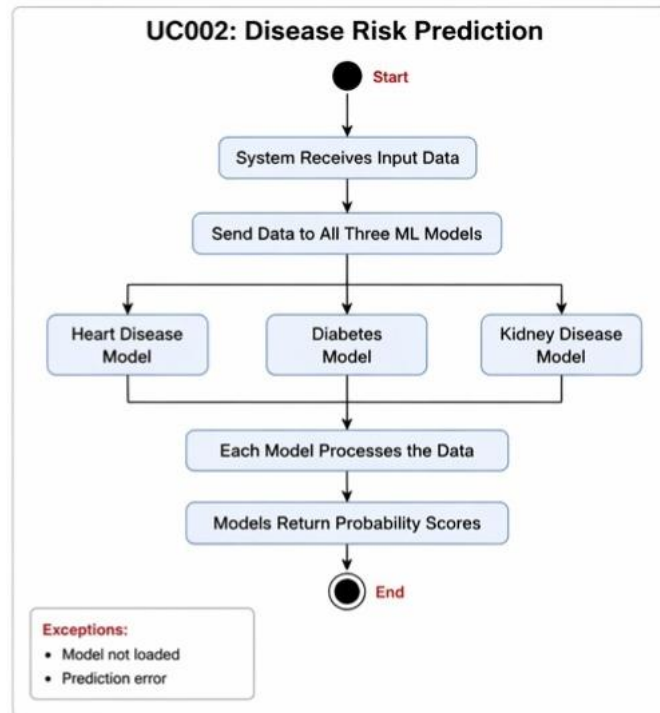
Activity diagram Use case 1:



3.1.2 Use case 2

ID	UC002
Name	Disease Risk Prediction
Description	This use case describes how the system predicts disease risks using ML models.
Requirement(s)	FR002
Actor(s)	System
Precondition	Valid patient data must be available
Postcondition	Risk probabilities are generated
Basic Flow	<i>Basic Flow</i> I. System receives input data II. Data is sent to all three ML models III. Each model processes the data IV. Models return probability scores <i>Alternative Flow</i> User may skip optional fields or enter invalid data, in which case the system shows an error message. The user can also reset the form to clear all inputs <i>Exceptions</i> 1. Model not loaded 2. Prediction error

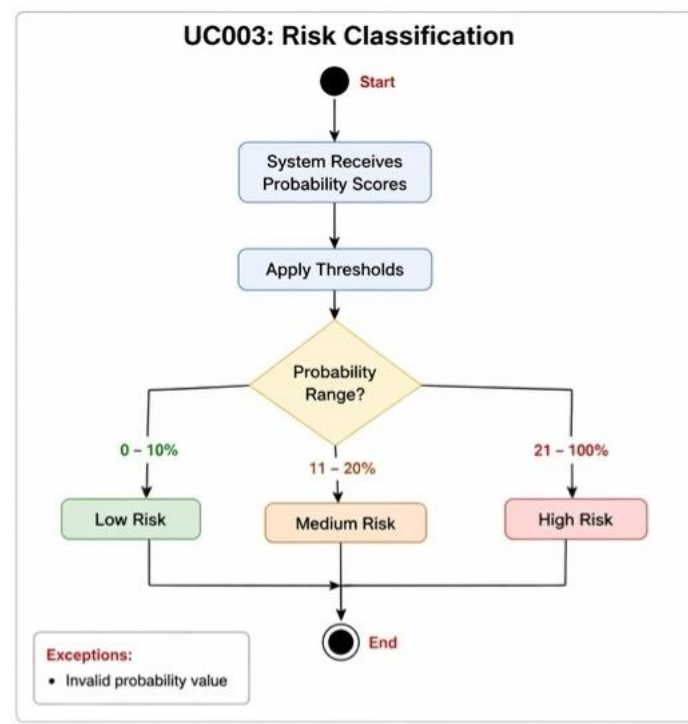
Activity diagram Use case 2:



3.1.3 Use case 3

ID	UC003
Name	Risk Classification
Description	This use case converts probabilities into risk levels.
Requirement(s)	FR003
Actor(s)	System
Precondition	Probability scores must be available
Postcondition	Risk categories are assigned
Basic Flow	Basic Flow ● System receives probability scores ● System applies thresholds ● Assigns Low, Medium, High risk Alternative Flow For boundary values, the system applies predefined rules. Risk categories or visualization may be updated in future versions. Exceptions ● Invalid probability value

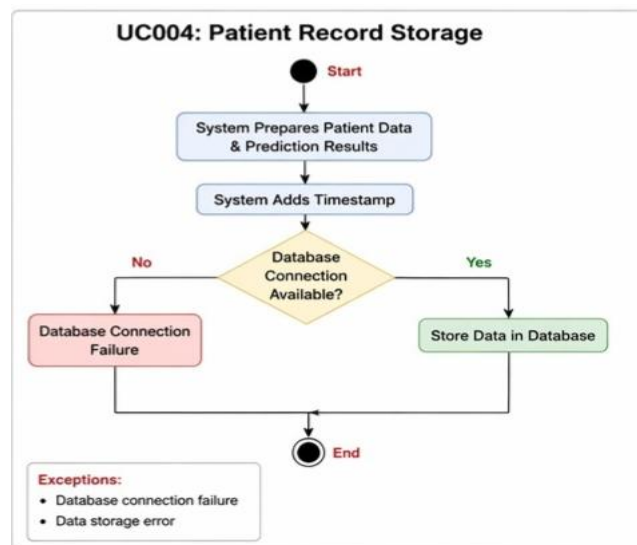
Activity diagram use case 3:



3.1.4 Use Case 4

ID	UC004
Name	Patient Record Storage
Description	This use case stores patient data and prediction results in the database.
Requirement(s)	FR004
Actor(s)	System / Admin
Precondition	Prediction must be completed
Postcondition	Record saved in database
Basic Flow	<i>Basic Flow</i> 1. System prepares patient data 2. System adds timestamp 3. System stores data in database <i>Alternative Flow</i> If duplicate data is detected, the system may update or create a new record. In case of temporary issues, data can be saved later. <i>Exceptions</i> I. Database connection failure II. Data storage error

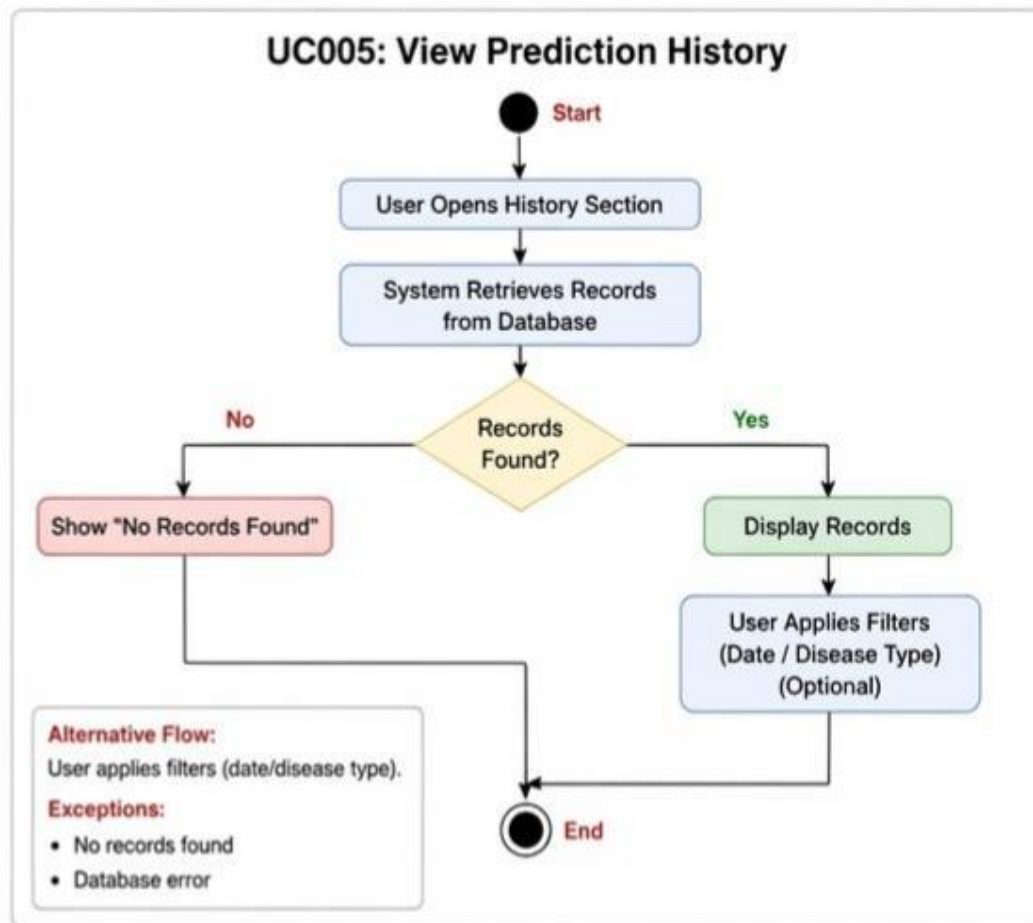
Activity diagram Use case 4:



3.1.5 Use case 5

ID	UC005
Name	View Prediction History
Description	This use case allows viewing past prediction records.
Requirement(s)	FR005
Actor(s)	Admin / Authorized User
Precondition	Records must exist in database
Postcondition	History is displayed
Basic Flow	<i>Basic Flow</i> 1. User opens history section 2. System retrieves records from database 3. System displays records <i>Alternative Flow</i> User can filter records by date or disease type or search by ID. The system may display sorted or limited records. <i>Exceptions</i> ● No records found ● Database error

Activity diagram Use case 5:



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